Robot Learning

Lecture 8 Bandits and the Exploration-Exploitation dilemma

News

- CPD
- Lecture on Dec 23
- Project assistance in Jan? (1.5h+?)

Outline

- Introduction
- Exploration and Exploitation
  - Simple naïve exploration (ϵ-greedy)
  - Optimistic approaches
  - Probability matching & Information Value
- Bandits
  - Multi-armed
  - Contextual
- Back to MDPs

Introduction

Exploration-Exploitation Dilemma

- Online decision-making involves a fundamental choice:
  - Exploitation - Make the best decision given current information
  - Exploration - Gather more information
- The best long-term strategy may involve short-term sacrifices
- Gather enough information to make the best overall decisions

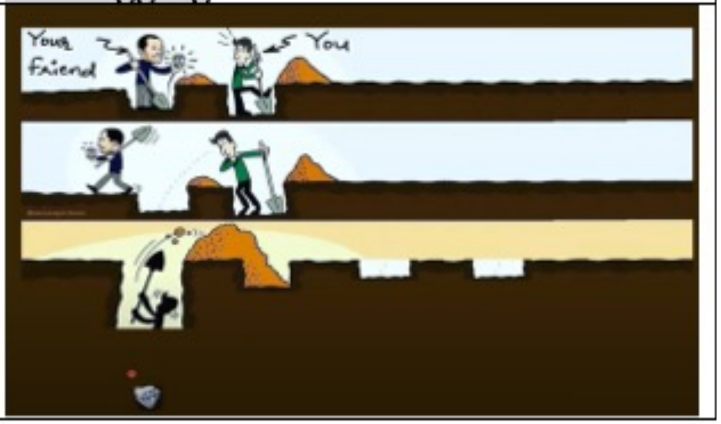
Examples

- Restaurant Selection
  - Exploitation - Go to your favourite restaurant
  - Exploration - Try a new restaurant
- Holiday planning
  - Exploitation – The camping site you go to since you are born
  - Exploration – Hitchhike and follow the flow
- Game Playing
  - Exploitation - Play the move you believe is best
  - Exploration - Play an experimental move

Exploration/Exploitation visualized

Exploration/Exploitation visualized

Diamond present where your friend was digging



Exploration/Exploitation visualized

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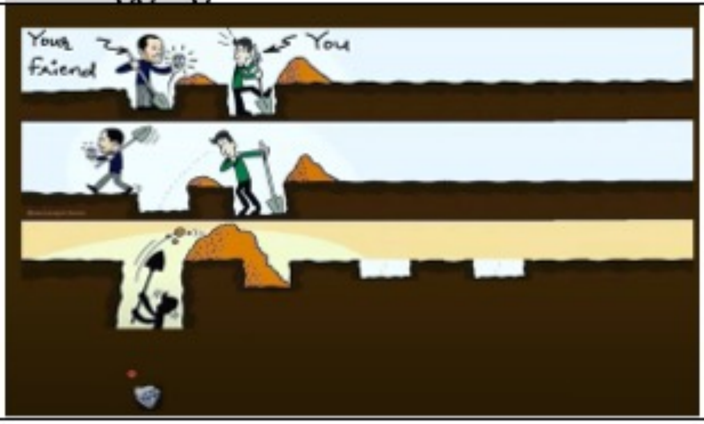
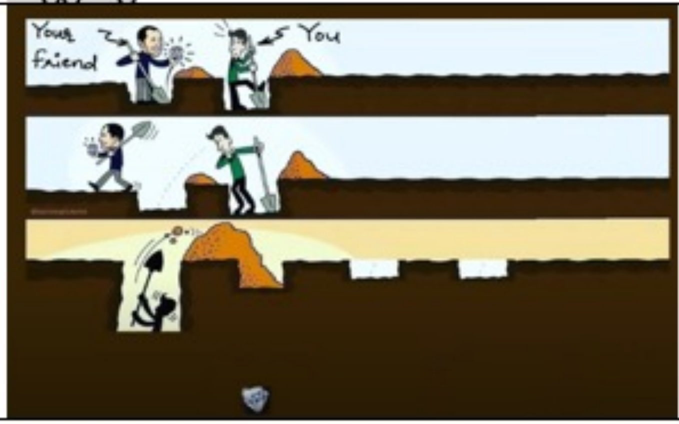
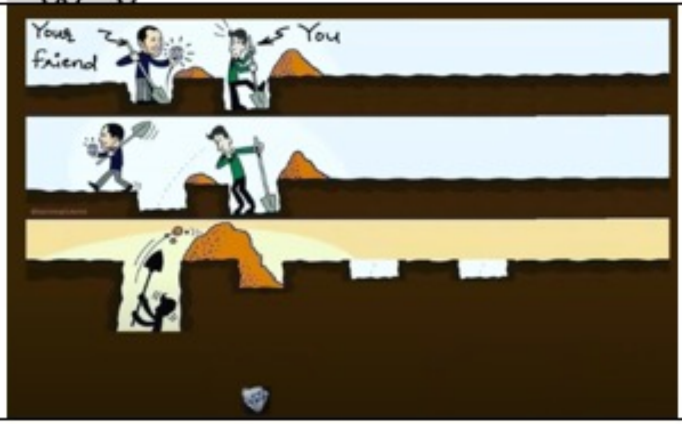
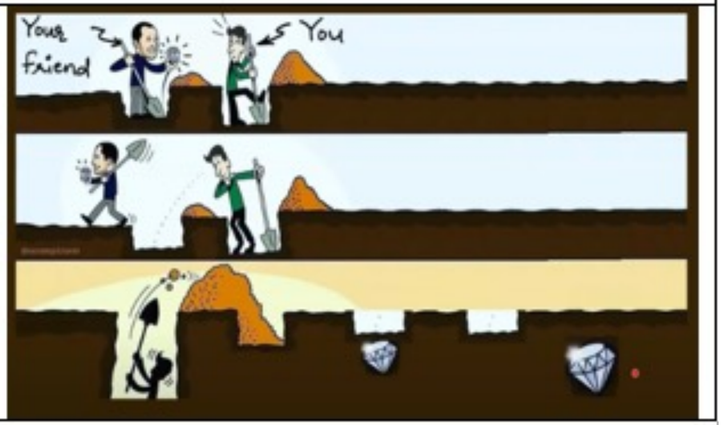
	

Exploration/Exploitation visualized

Diamond present where your friend was digging	Diamond present where you were digging	Present somewhere else
		

Exploration/Exploitation visualized

Greedy Action with probability 1-ε (Exploitation)

Exploration with probability ε

0<ε<1

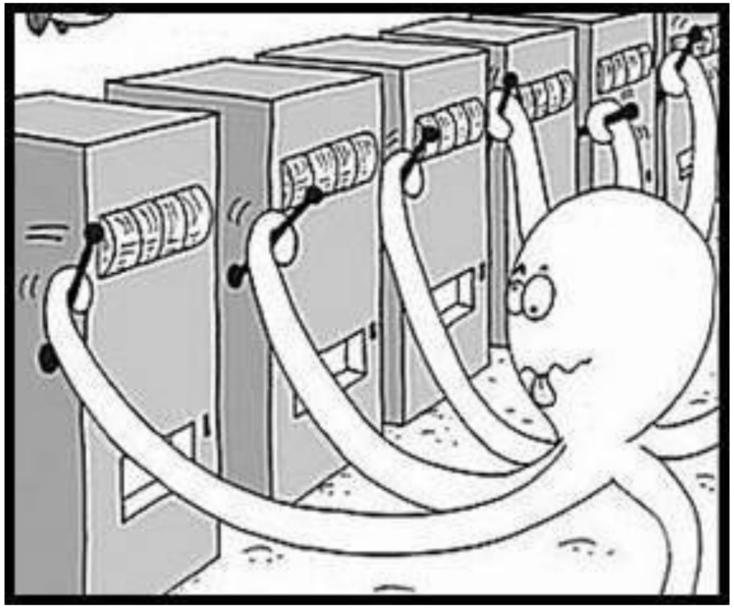
Principles

- Random Exploration
  - Add noise to greedy policy (e.g. ϵ-greedy)
- Optimism in the Face of Uncertainty
  - Estimate uncertainty on value
  - Prefer to explore states/actions with highest uncertainty
- Information State Search
  - Consider agent's information (history) as part of its state
  - Lookahead to see how information helps rewards

Bandits

Multi-Armed Bandit

- A multi-armed bandit is a tuple $\langle \mathcal{A}, \mathcal{R} \rangle$
  - $\mathcal{A}$ is a known set of m actions (or "arms")
  - $\mathcal{R}^a(r) = P(r|a)$ is an unknown probability distribution over rewards
- At each step t the agent selects an action $a \in \mathcal{A}$
- The environment generates a reward $r_t \sim \mathcal{R}^{a_t}$
- The goal is to maximise cumulative reward $\sum_{\tau=1}^{t} r_\tau$



Regret

Instead of $Q(s, a)$, because we have a single state

- The action-value is the mean reward for action a

$$Q(a) = \mathbb{E}[r|a]$$

- The optimal value V^* is

$$V^* = Q(a^*) = \max_{a \in \mathcal{A}} Q(a)$$

- The regret is the opportunity loss for one step

$$I_t = \mathbb{E}[V^* - Q(a_t)]$$

- The total regret is the total opportunity loss

$$L_t = \mathbb{E}\left[\sum_{\tau=1}^{t} V^* - Q(a_t)\right]$$

- Maximise cumulative reward ≡ minimise total regret

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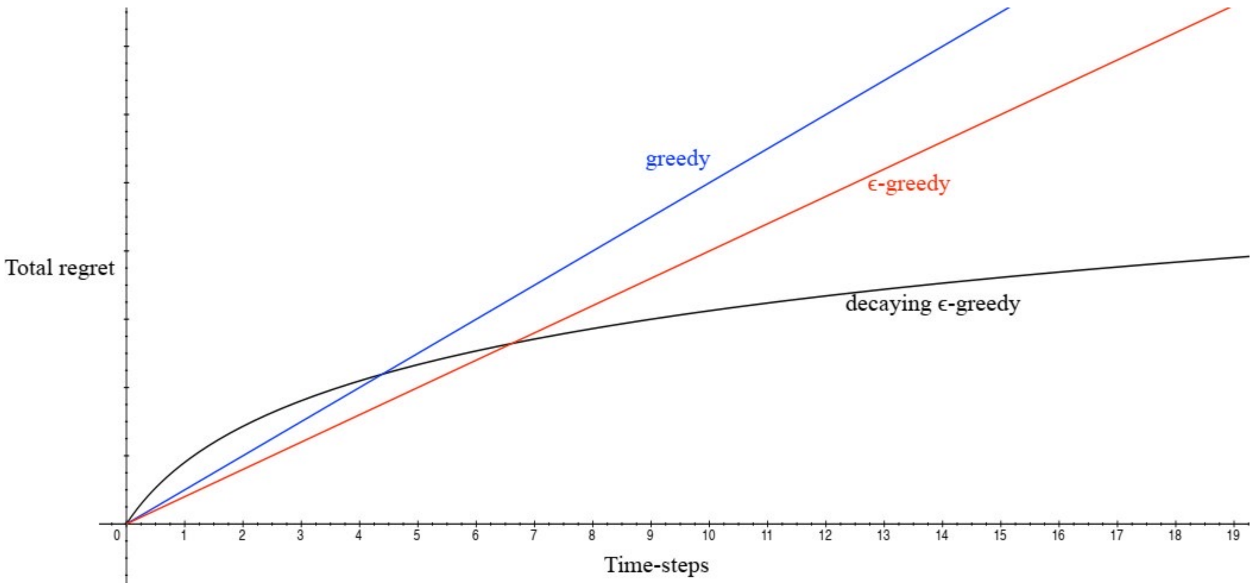
Counting Regret

- The **count $\mathbf{N_t}(\boldsymbol{a})$** is the expected number of selections for action a (how many times we pull that specific lever)
- The **gap** (for action a) $\boldsymbol{\Delta_a} = V^* - Q(a)$ is the difference in value between action a and optimal action a^* (zero for optimal action)
- Regret is a **function of gaps and the counts**

$$L_t = \mathbb{E}\left[\sum_{\tau=1}^{t} V^* - Q(a_\tau)\right] = \sum_{a \in \mathcal{A}} \mathbb{E}[N_t(a)](V^* - Q(a_\tau)) = \sum_{a \in \mathcal{A}} \mathbb{E}[N_t(a)]\Delta_\mathrm{a}$$

- A good algorithm ensures small counts for large gaps
- Problem: gaps are not known! – because I don't know the optimal V^*

Linear or Sublinear Regret



- If an algorithm forever explores it will have linear total regret
- If an algorithm never explores it will have linear total regret
- Is it possible to achieve sublinear total regret?

Exploration Strategies

Greedy Algorithms

- We consider algorithms that estimate $\hat{Q}_t(a) \approx Q(a)$
- Estimate the value of each action by Monte-Carlo evaluation

$$\hat{Q}_t(a) = \frac{1}{N_t(a)} \sum_{\tau} r_\tau \mathbf{1}(a_\tau; a)$$

- The greedy algorithm selects action with highest value

$$a_t^* = arg \max_{a \in \mathcal{A}} \hat{Q}_t(a)$$

- Greedy can lock onto a suboptimal action forever (could be mitigated with a very optimistic optimization)

Greedy has linear total regret

ϵ-greedy Algorithms

- The ϵ-greedy algorithm continues to explore forever
  - With probability $1 - \epsilon$ select $a = \max_{a \in \mathcal{A}} \hat{Q}(a)$
  - With probability ϵ select a random action
- Constant ϵ ensures minimum total regret L_t (lower bound)

$$L_t \geq \frac{\epsilon}{|\mathcal{A}|} \sum_{a \in \mathcal{A}} \Delta_a$$

Action space size

ϵ-greedy has linear total regret

Decaying ϵ_t-greedy Algorithms

- Pick a decay schedule for $\epsilon_1, \epsilon_2, \ldots$
- Consider the following schedule

$$c > 0$$

$$d = \min_{a|\Delta_a > 0} \Delta_i$$

$$\epsilon_t = \min\left\{1, \frac{c|\mathcal{A}|}{d^2 t}\right\}$$

Ensures convergence

- Decaying ϵ_t-greedy has logarithmic asymptotic total regret
- Unfortunately, schedule requires advance knowledge of gaps
- Goal: find an algorithm with sublinear regret for any multi-armed bandit (without knowledge of R)

Lower Bound

- The performance of any algorithm is determined by similarity between optimal arm and other arms
- Hard problems have similar-looking arms with different means
- This is described formally by the gap Δ_a and the similarity in distributions $KL(\mathcal{R}^a || \mathcal{R}^{a^*})$

Lower Bound

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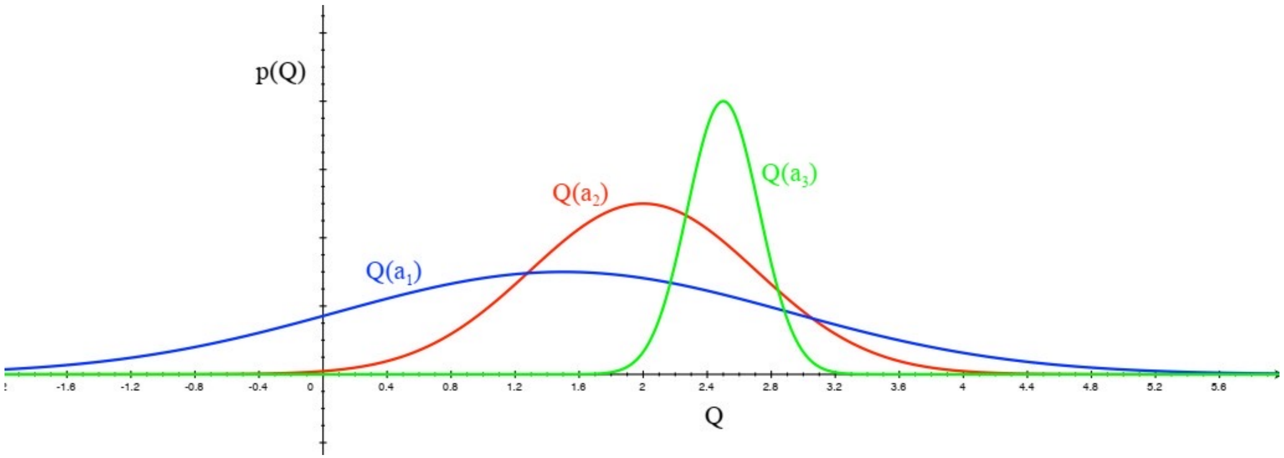
Theorem (Lai and Robbins)

Asymptotic total regret is at least logarithmic in the number of steps

$$\lim_{t \to \infty} L_t \geq \log t \sum_{a | \Delta_a > 0} \frac{\Delta_a}{KL(\mathcal{R}^a || \mathcal{R}^{a^*})}$$

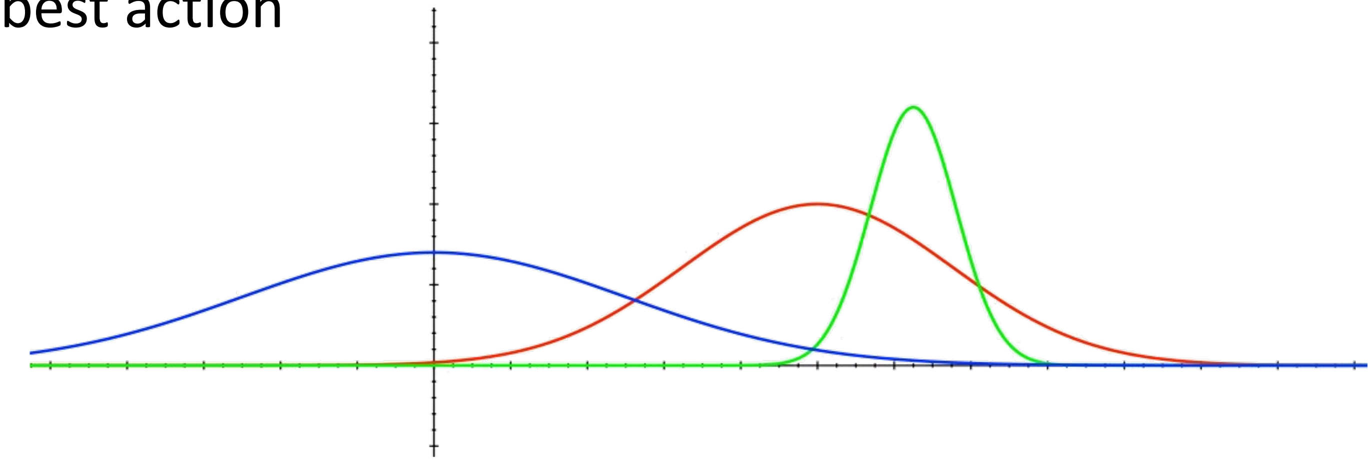
Optimism in the Face of Uncertainty (I)

- Which action should we pick?
- The more uncertain we are about an action-value, the more important it is to explore that action
- It could turn out to be the best action



Optimism in the Face of Uncertainty (II)

- After picking blue action
- We are less uncertain about the value
- And more likely to pick another action
- Until we home in on best action



Upper Confidence Bounds

- Estimate an upper confidence $\hat{U}_t(a)$ for each action value
- Such that $Q(a) \leq \hat{Q}_t(a) + \hat{U}_t(a)$ with high probability
- This depends on the number of times $N(a)$ has been selected
  - Small $N_t(a) \Rightarrow$ large $\hat{U}_t(a)$ (estimated value is uncertain)
  - Large $N_t(a) \Rightarrow$ small $\hat{U}_t(a)$ (estimated value is accurate)
- Select action maximising Upper Confidence Bound (UCB)

$$a_t = arg \max_{a \in \mathcal{A}} \hat{Q}_t(a) + \hat{U}_t(a)$$

Hoeffding's Inequality

Theorem (Hoeffding's Inequality)

Let $X_1, \dots, X_t$ be i.i.d. random variables in $[0,1]$ and let $\bar{X}_t = \frac{1}{\tau}\sum_{\tau=1}^{t} X_t$ the sample mean, then

$$P(\mathbb{E}[X] > \bar{X}_t + u) \leq e^{-2tu^2}$$

We will apply Hoeffding's Inequality to rewards of the bandit conditioned on selecting action a

$$P\big(Q(a) > \hat{Q}_t(a) + U_t(a)\big) \leq e^{-2N_t(a)U_t(a)^2}$$

Calculating Upper Confidence Bounds

- Pick a probability p that true value exceeds UCB
- Now solve for $U_t(a)$

$$e^{-2N_t(a)U_t(a)^2} = p$$

$$U_t(a) = \sqrt{\frac{-\log p}{2N_t(a)}}$$

- Reduce p as we observe more rewards, e.g. $p = t^{-4}$
- Ensures we select optimal action as $t \to \infty$

$$U_t(a) = \sqrt{\frac{2\log t}{N_t(a)}}$$

UCB1

- This leads to the UCB1 algorithm

$$a_t = \arg\max_{a \in \mathcal{A}} Q(a) + \sqrt{\frac{2\log t}{N_t(a)}}$$

- The UCB algorithm achieves logarithmic asymptotic total regret

UCB vs. ϵ- greedy on 10- armed Bandit

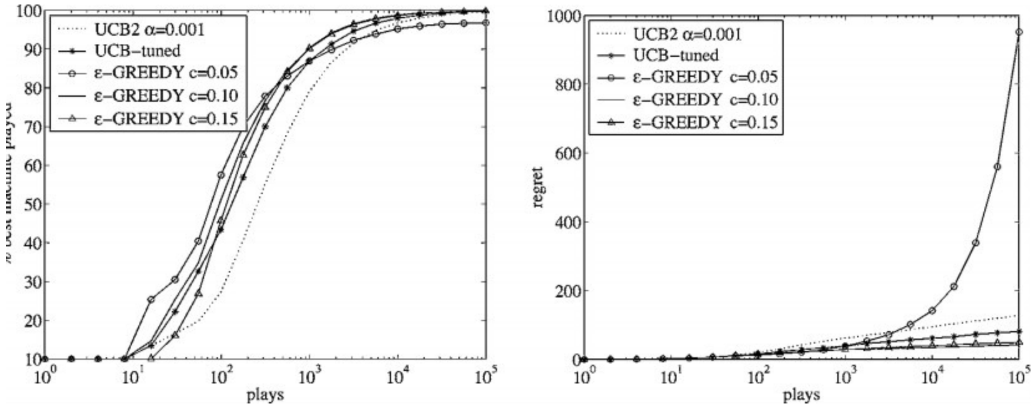
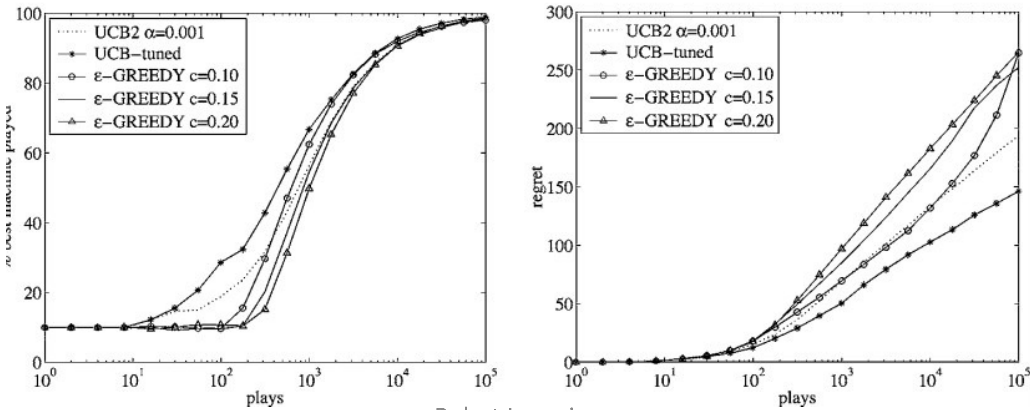


Figure 9. Comparison on distribution 11 (10 machines with parameters 0.9, 0.6, . . . , 0.6).



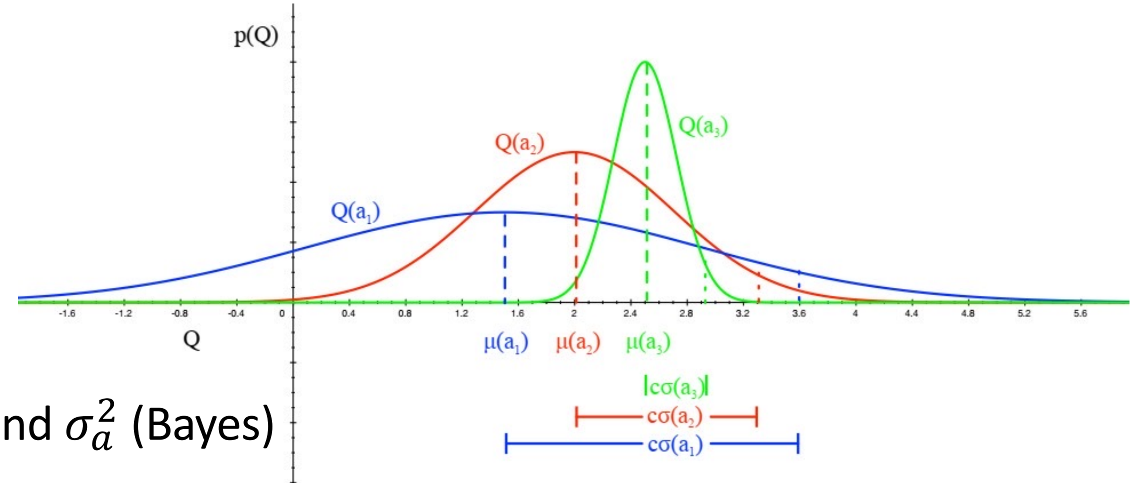
Bayesian Bandits

- So far, no assumptions about the reward distribution $\mathcal{R}$
  - Except bounds on rewards
- Bayesian bandits exploit prior knowledge of rewards $P(\mathcal{R})$
- They compute posterior distribution of rewards $P(\mathcal{R}|\mathrm{h_t})$
  - where $h_t = a_1, r_1; \dots; a_{t-1} r_{t-1}$ is the history
- Use posterior to guide exploration
  - Upper confidence bounds (Bayesian UCB)
  - Probability matching (Thompson sampling)
- Better performance if prior knowledge is accurate

Bayesian UCB Example - Independent Gaussians

Assume reward distribution is Gaussian

$$\mathcal{R}_a(r) = \mathcal{N}(r; \mu_a, \sigma_a^2)$$



- Compute Gaussian posterior over μ_a and σ_a^2 (Bayes)

$$P(\mu_a, \sigma_a^2 | h_t) \propto P(\mu_a, \sigma_a^2) \prod_{t | a_t = a} \mathcal{N}(r_t; \mu_a, \sigma_a^2)$$

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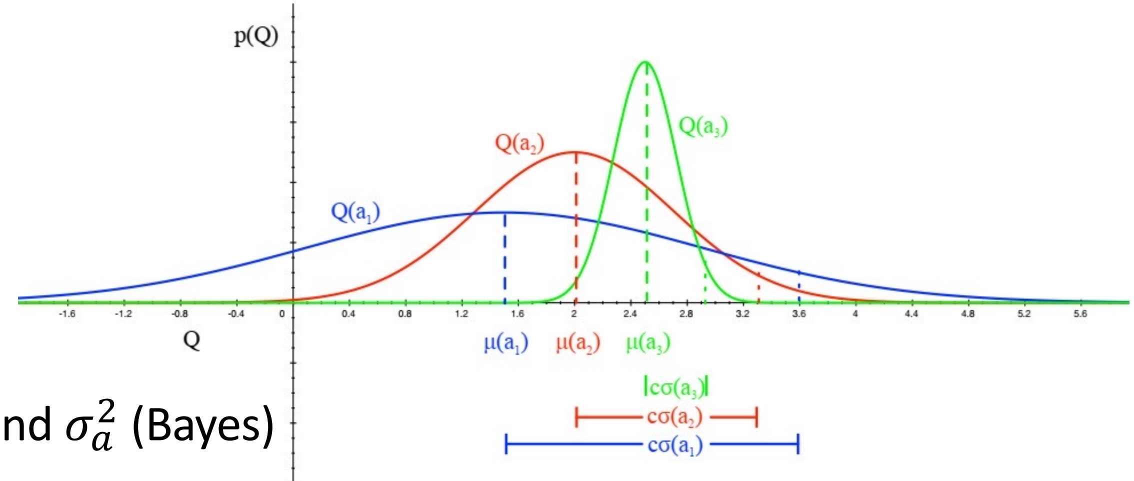

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$$P(\mu_a, \sigma_a^2 | h_t) \propto P(\mu_a, \sigma_a^2) \boxed{\prod_{t|a_t=a} \mathcal{N}(r_t; \mu_a, \sigma_a^2)}$$

Prior distribution

Likelihood of experiencing a reward r_t given the distribution

Bayesian UCB Example - Independent Gaussians

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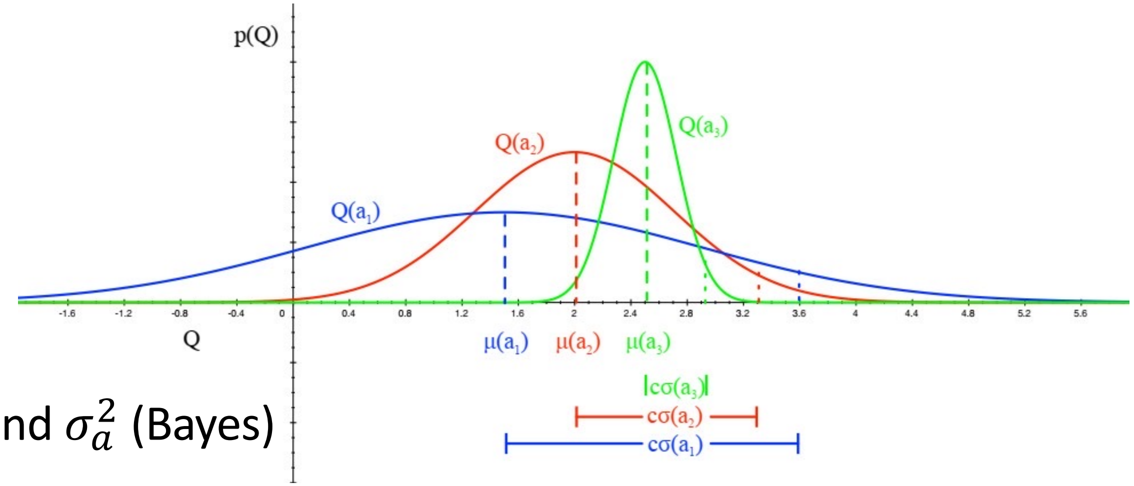

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- Pick action that maximises standard deviation of $Q(a)$

$$a_t = arg \max \mu_a + c\sigma_a / \sqrt{N(a)}$$

Probability Matching

- Probability matching selects action a according to probability that a is the optimal action

$$\pi(a|h_t) = P\left(Q(a) = \max_{a'} Q(a')|h_t\right)$$

- Probability matching is optimistic in the face of uncertainty
  - Uncertain actions have higher probability of being max
- Can be difficult to compute analytically from posterior
  - Needs to be approximated through sampling

Thompson Sampling

- Thompson sampling implements probability matching
  $$\pi(a|h_t) = \mathbb{E}_{\mathcal{R}|h_t}\left[\mathbf{1}\left(Q(a); arg \max_{a' \in \mathcal{A}} Q(a')\right)|h_t\right]$$
- Use Bayes law to compute posterior distribution $P(\mathcal{R}|h_t)$
- Sample a reward distribution $\mathcal{R}$ from posterior
- Compute action-value function $Q(a) = \mathbb{E}[\mathcal{R}_a]$
- Select action maximising value on sample $a_t = arg \max_{a' \in \mathcal{A}} Q(a)$
- Thompson sampling achieves Lai and Robbins lower bound!

Information State

Value of Information

- Exploration is useful because it gains information
- Can we quantify the value of information?
  - How much reward a decision-maker would be prepared to pay in order to have that information, prior to making a decision
  - Long-term reward after getting information - immediate reward
- Information gain is higher in uncertain situations
- Therefore, it makes sense to explore uncertain situations more
- If we know value of information, we can trade-off exploration and exploitation optimally

Information State Space

- We have viewed bandits as one-step decision-making problems
- Can also view as sequential decision-making problems
- At each step there is an information state $\tilde{s}$
  - $\tilde{s}$ is a statistic of the history, i.e. $\tilde{s} = f(h_t)$
  - summarizes all information accumulated so far
- Each action a causes a transition to a new information state $\tilde{s}'$ (and adds information) with probability $\tilde{P}^{a}_{\tilde{s},\tilde{s}'}$
- Defines an MDP $\tilde{\mathcal{M}}$ in augmented information state space

$$\tilde{\mathcal{M}} = \langle \tilde{\mathcal{S}}, \mathcal{A}, \tilde{\mathcal{P}}, \mathcal{R}, \gamma \rangle$$

Example - Bernoulli Bandits

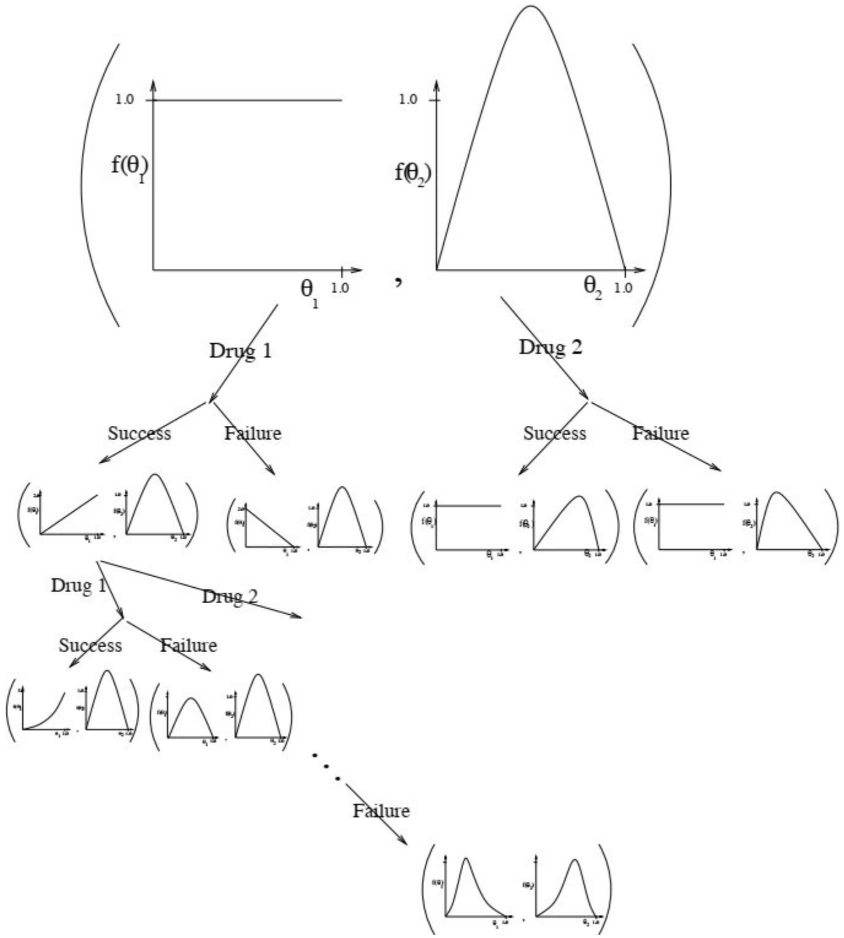
- Consider a Bernoulli bandit, such that $R_a = B(\mu_a)$ (e.g. win or lose a game with probability μ_a)
- Want to find which arm has the highest μ_a
- The information state is $\tilde{s} = \langle \alpha, \beta \rangle$
  - α_a counts the pulls of arm a where reward was 0
  - β_a counts the pulls of arm a where reward was 1

Solving Information State Space Bandits

- We now have an infinite MDP over information states that can be solved by reinforcement learning
- Model-free reinforcement learning
  - e.g. Q-learning (Duff, 1994)
- Bayesian model-based reinforcement learning
  - e.g. Gittins indices (Gittins, 1979)
  - This approach is known as Bayes-adaptive RL
  - Finds Bayes-optimal exploration/exploitation trade-off with respect to prior distribution

Bayes-Adaptive Bernoulli Bandits

- Start with $Beta(\alpha_a, \beta_a)$ prior over reward function R_a
- Each time a is selected, update posterior for R_a
  - $Beta(\alpha_a + 1, \beta_a)$ if r = 0
  - $Beta(\alpha_a, \beta_a + 1)$ if r = 1
- This defines transition function $\tilde{P}$ for the Bayes-adaptive MDP
- Information state $\langle \alpha, \beta \rangle$ corresponds to reward model $Beta(\alpha, \beta)$
- Each state transition corresponds to a Bayesian model update



Gittins Indices for Bernoulli Bandits

- Bayes-adaptive MDP can be solved by dynamic programming
- The solution is known as the Gittins index
- Exact solution to Bayes-adaptive MDP is typically intractable
  - Information state space is too large
- More recent idea: apply simulation-based search (Guez et al. 2012)
  - Forward search in information state space
  - Using simulations from current information state

Contextual Bandits

Contextual Bandits

- A contextual bandit is a tuple $\langle \mathcal{A}, \color{red}{\mathcal{S}}, \mathcal{R} \rangle$
- $\color{red}{\mathcal{S}} = P(s)$ is an unknown distribution over states (contexts)
- $\mathcal{R}_{\color{red}{s}}^{a}(r) = P(r|\color{red}{s}, a)$ is an unknown probability distribution over rewards
- At each step t
  - Environment generates state $s_t \sim \mathcal{S}$
  - Agent selects action $a_t \sim \mathcal{A}$
  - Environment generates reward $r_t \sim \mathcal{R}_{s_t}^{a_t}$

Linear Regression

- Action-value function is expected reward for state s and action a

$$Q(s,a) = \mathbb{E}[r|s,a]$$

- Estimate value function with a linear function approximator

$$Q_\theta(s,a) = \phi(s,a)^T\theta \approx Q(s,a)$$

- Estimate parameters by least squares regression

$$A_t = \sum_{\tau=1}^{t} \phi(s_\tau, a_\tau)\phi(s_\tau, a_\tau)^T$$

$$b_t = \sum_{\tau=1}^{t} \phi(s_\tau, a_\tau) r_\tau$$

$$\theta_t = A_t^{-1}\mathrm{b}_\mathrm{t}$$

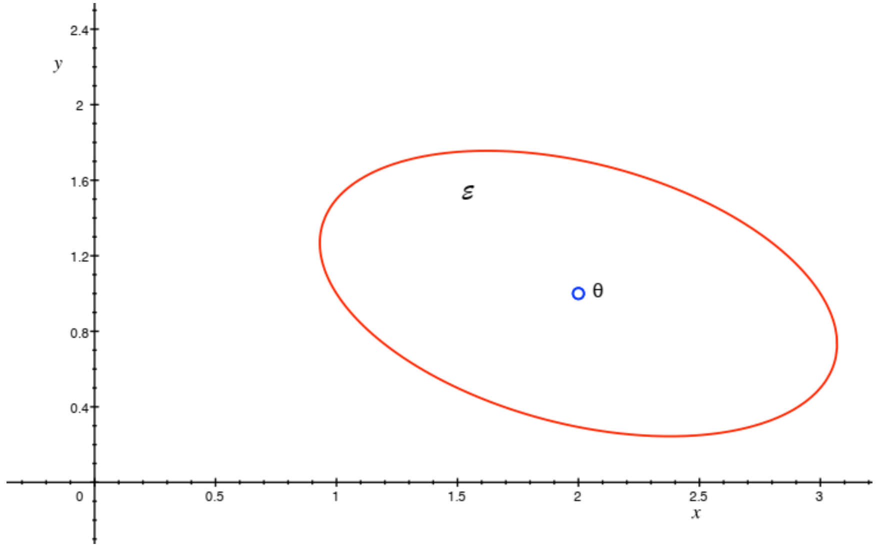
Linear Upper Confidence Bounds

- Least squares regression estimates the mean action-value $Q_\theta(s, a)$
- But it can also estimate the variance of the action-value $\sigma_\theta^2 = (s, a)$
  - i.e. the uncertainty due to parameter estimation error
- Add on a bonus for uncertainty, $U_\theta(s, a) = c\sigma$
  - i.e. define UCB to be c standard deviations above the mean

Geometric Interpretation

- Define confidence ellipsoid ε_t around parameters θ_t
- Such that ε_t includes true parameters θ^* with high probability
- Use this ellipsoid to estimate the uncertainty of action values
- Pick parameters within ellipsoid that maximise action value

$$arg \max_{\theta \in \epsilon} Q_{\theta}(s, a)$$



Calculating Linear Upper Confidence Bounds (LinUCB)

- For least squares regression, parameter covariance is A^{-1}
- Action-value is linear in features

$$Q_\theta(s,a) = \phi(s,a)^T \theta$$

- So, action-value variance is quadratic

$$\sigma_\theta^2(s,a) = \phi(s,a)^T A^{-1} \phi(s,a)$$

- Upper confidence bound is

$$Q_\theta(s,a) + c\sqrt{\phi(s,a)^T A^{-1} \phi(s,a)}$$

- Select action maximizing upper confidence bound

$$a_t = arg \max_{a \in \mathcal{A}} Q_\theta(s_t,a) + c\sqrt{\phi(s,a)^T A^{-1} \phi(s,a)}$$

Exploration-Exploitation in MDPs

Applying Exploration/Exploitation to MDPs

The same principles for exploration/exploitation apply to MDPs

- Naive Exploration
- Optimism in the Face of Uncertainty
- Probability Matching
- Information State Search

Upper Confidence Bounds - Model-Free RL (Policy iteration)

- Maximise UCB on action-value function $Q^{\pi}(s,a)$
$$a_t = arg\max_{a\in\mathcal{A}} Q(s_t,a) + U(s_t,a)$$
  - Estimate uncertainty in policy evaluation
  - Ignore uncertainty from policy improvement

- Maximise UCB on optimal action-value function $\mathrm{Q}^*(\mathrm{s},\mathrm{a})$
$$a_t = arg\max_{a\in\mathcal{A}} Q(s_t,a) + U_1(s_t,a) + U_2(s_t,a)$$
  - Estimate uncertainty in policy evaluation
  - Estimate uncertainty from policy improvement

Information State Search in MDPs

- MDPs can be augmented to include information state
- Now the augmented state is $\langle s, \tilde{s} \rangle$
  - s is original state within MDP
  - $\tilde{s}$ is a statistic of the history
- Each action *a* causes a transition
  - to a new state s' with probability $\mathcal{P}^{a}_{s,s'}$
  - to a new information state $\tilde{s}'$
- Defines MDP $\widetilde{\mathcal{M}}$ in augmented information state space
$$\widetilde{\mathcal{M}} = \langle \tilde{\mathcal{S}}, \mathcal{A}, \tilde{\mathcal{P}}, \mathcal{R}, \gamma \rangle$$
- Posterior distribution over MDP model is an information state
$$\tilde{s} = P(\mathcal{P}, \mathcal{R} | h_t)$$
- Solve this MDP to find optimal exploration/exploitation trade-off (w.r.t. prior)

Wrap-up

Take home messages

- A selection of principles for exploration/exploitation
  - Naive methods (ϵ-greedy)
  - Upper confidence bounds
  - Probability matching
  - Information state search
- Principles developed in bandit setting but also apply to MDP setting

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Coming up

- Sim-to-real transfer
  - How can I manage to learn in simulation and transfer a policy in the real world
- Imitation Learning
- Research Topics
- Thesis ideas